

ASSEMBLYAI VOICE AGENT HACKATHON • LABLAB.AI

Heard.

The drive-thru AI that doesn't make up orders.

Hears the driver through engine noise and back-seat chatter, builds a verified order from what it actually heard, and keeps a person one sentence away.

THE PROBLEM

Every viral AI drive-thru failure was a **hearing** failure.

McDonald's · June 2024

Ended its IBM AI drive-thru test and pulled it from 100+ restaurants after viral misheard orders, like nine sweet teas instead of one.

Taco Bell · 2025

Slowed its AI rollout after glitches and trolling, including a caller who ordered 18,000 waters just to reach a human.

The hardest place to listen

Engines, traffic, kids in the back seat, a cheap speaker a metre away, and guests who change their mind mid-sentence.

Sources: CNBC, Restaurant Dive (Jun 2024); Jalopnik, AOL (2025). Full links in the repo: docs/sources.md

THE SOLUTION

Hear the driver. Verify the order. Keep it human.

1 • Hears

AssemblyAI Universal-3.6 Pro with voice focus (AssemblyAI lists it for drive-thru speakers) and the menu as key terms, so "Cluckwich" survives the engine noise.

2 • Verifies

The AI runs the conversation but never writes the order. A deterministic parser builds it from the transcript; a tested engine owns every item, price and total.

3 • Stays human

"Can I talk to a person?" works in one sentence. Spanish, English or both. Tickets go straight to the kitchen screen.

“Two Spicy Cluckwiches, **actually** make one of those a Stackhouse Double, **no pickles**, and a large chocolate Frostee. **Wait, scratch the Frostee.**”

Your order

2 items

1

Spicy Cluckwich

\$5.79

1

Stackhouse Double

\$6.29

NO PICKLES

Subtotal

\$12.08

Tax

\$0.97

Total

\$13.05

Order confidence

Every line traces to the driver's words. Nothing is sent until it's read back.

4/5 checks

✓ HEARD YOU

5 sentences

"Yes, that's right."

✓ ORDER BUILT

2 items

from your words by the parser, not by the...

✓ CORRECTIONS

2 applied

"scratch the Frosty."

✓ IGNORED

1 ignored

1 suspicious quantity

... READ BACK

waiting for "yes"

nothing goes to the kitchen until the gues...

ORDER CONFIDENCE · 5 OF 5

✓ Heard 5 sentences from the driver

✓ Built from the driver's own words, not by the AI

✓ 2 corrections applied, in order

last one: "scratch the Frosty."

✓ Ignored 1 suspicious quantity

✓ Read back and confirmed by the driver

#988

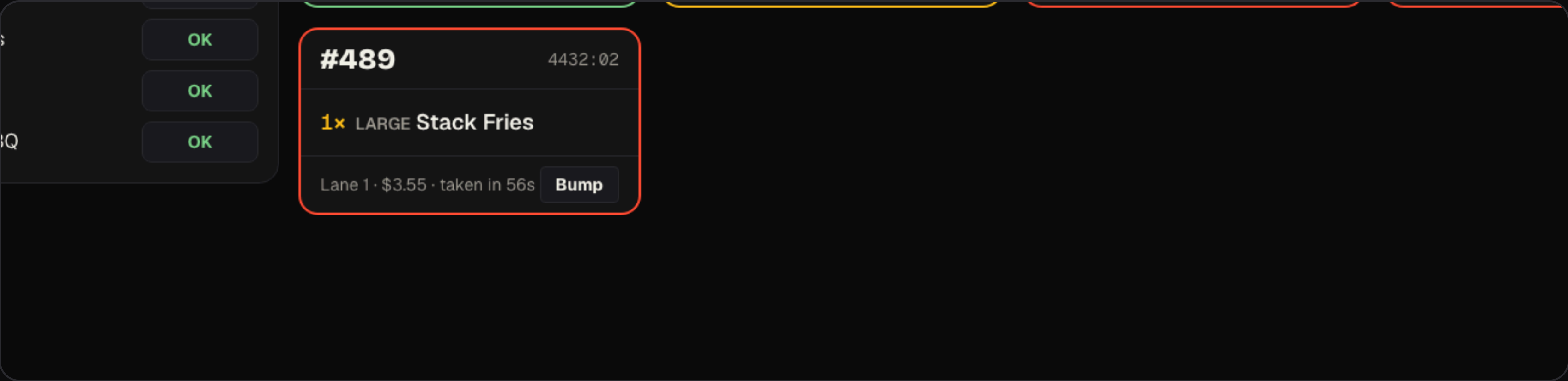
\$13.05

Sent to the kitchen · please pull forward

Click anywhere to continue

A real Voice Agent session: the order comes out exactly right, and all five confidence checks pass before the ticket fires. Every change shows the words it came from and the rule that applied.

From the speaker to the kitchen, with no keyboard.



NO PICKLES

in red on the ticket, the moment the guest confirms.

“Oh wait, a cola” after the total updates the same ticket

Frostee machine down? Mark it, and Heard stops selling it immediately

THE PROOF • NOISE BENCHMARK

Road noise as loud as the driver:

13%

of orders exactly right with generic speech-to-text

99%

with Heard. 96–100% in every condition.



24 order lines • 6 English accents • streamed in real time through AssemblyAI • mean of 2–3 sessions per cell • same parser as the live agent. Synthetic voices and noise; reproducible with `npm run bench`.

SAME AUDIO, TWO EARS

Made-up brand names are exactly what generic STT gets wrong.

SAID Let me get a Spicy Cluckwich meal, large, with a Sweet Tea.

GENERIC ✗ let me get a spicy clockwich meal with a sweet tea?

HEARD ✔ Let me get a Spicy Cluckwich meal, large, with a sweet tea.

SAID Can I get ten Cluck Bites with honey mustard and ranch.

GENERIC ✗ Can I get 10 plug bites with honey mustard hand ranch

HEARD ✔ Can I get 10 Cluck Bites with honey mustard and ranch?

SAID I'll do the Smokestack BBQ with extra cheese and add bacon.

GENERIC ✗ I'll do theq with extra cheese and Adbacon

HEARD ✔ I'll do the Smokestack BBQ with extra cheese and add bacon.

SAID Two Stack Packs, one with the mini burger and one with Cluck Bites.

GENERIC ✗ 2 stack packs one with the mini burger and one with club bites.

HEARD ✔ 2 Stack Packs, one with the mini burger and one with Cluck Bites.

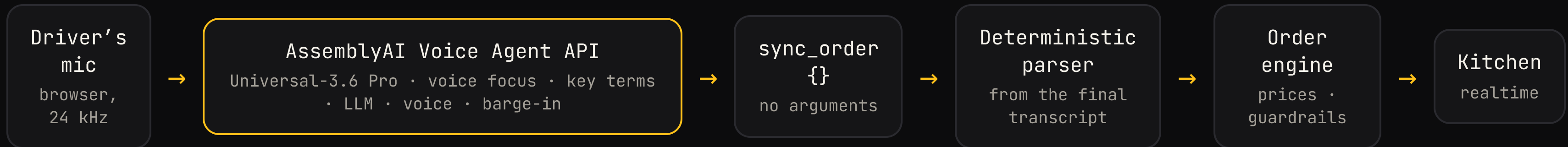
"Cluck Bites" → "plug bites"

"Smokestack BBQ" → "theq"

"Spicy Cluckwiches" → "USPYC Club
witches"

In the live demo, judges see this side
by side for their own voice: **Hear it both
ways.**

The model runs the conversation. Code owns the order.



Why no tool arguments?

Live testing showed the managed LLM returned an empty reply whenever a turn needed several changes. A parameterless tool is called reliably, and the order becomes deterministic and testable.

Engineering

- ▶ 104 automated tests
- ▶ Scripted scenarios verified against the live agent
- ▶ Median reply 1.25 s, p90 1.68 s
- ▶ Single-use browser tokens; MIT-licensed

Built so everyone gets understood the first time.

Nine sweet teas instead of one	Quantities parsed deterministically, read back before sending
18,000 waters	Quantity caps with a friendly double-check
Trapped with a bot	A person is one sentence away; crew alarm on the kitchen screen
Invented items or prices	Impossible: the AI never writes the order
Pushy upselling	At most one offer per order

Everyone, any car

- ▶ Tested across 6 English accents
- ▶ Spanish, English or both in one sentence
- ▶ Every order shown on screen as it's heard, for hard-of-hearing drivers too
- ▶ Asks instead of guessing a size or flavor

The hearing layer for voice AI in loud places.

Who pays

Quick-service chains and franchisees. Proposed model: a monthly subscription per drive-thru lane.

Why switch: accuracy you can measure before you deploy, orders that are never invented, a human always one sentence away, and bilingual service.

Landscape

Presto, SoundHound and Wendy's FreshAI (with Google Cloud) show the demand. Heard's difference: hearing-first, deterministic orders, and published accuracy.

Illustrative only: 500 drive-thru orders a day with 2% fewer remakes is 10 orders a day, about 3,650 a year per store.

Next

- ▶ Pharmacy drive-thru windows, where a misheard name or drug is a safety risk
- ▶ Pilot recordings from a real lane
- ▶ Point-of-sale integration
- ▶ More languages